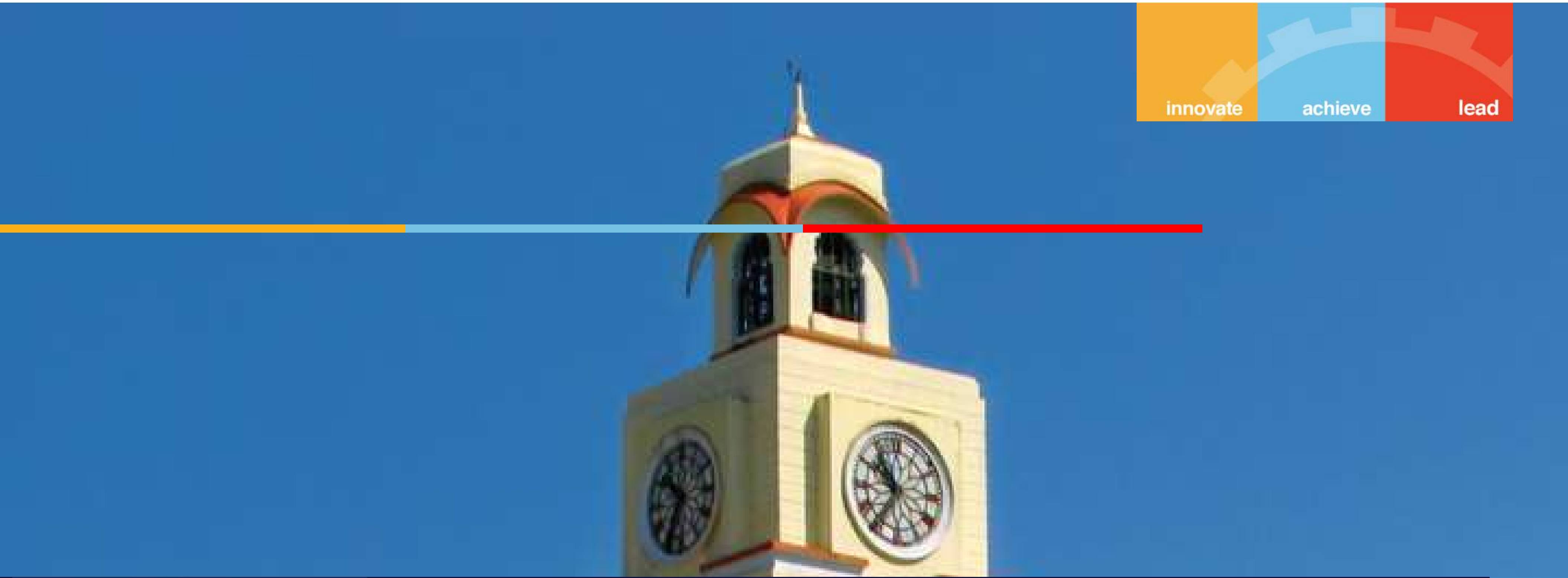


innovate achieve lead

BITS Pilani
Pilani Campus

CE F230
Civil Engineering Materials

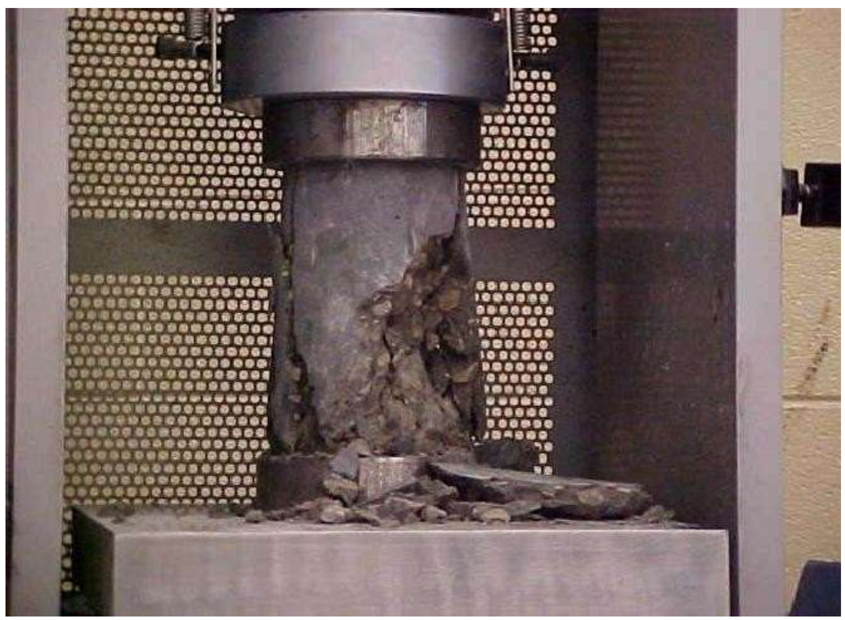
Testing of Hardened Concrete

Need for "Standard" Tests

- Why testing hardened concrete properties?
  - To provide information on quality of concrete
- Variables affects testing results
  - Specimen geometry, specimen preparation, moisture content, temperature, loading rate, and type of testing machine and loading fixture
  - Different test procedures yields different results
- "Standard" test methods are proposed to minimize confusion that would result if everyone were to use different test procedures.

Test on Compressive Strength

	ASTM	British Standard
Specimen geometry	Cylinder 150 mm dia., 300 mm long	Cube 150 mm
Curing temperature	23 ± 2^{o}C	20 ± 2^{o}C
Loading rate	0.15 to 0.35 MPa/s	0.2 to 0.4 MPa/s
End preparation	Required	No need



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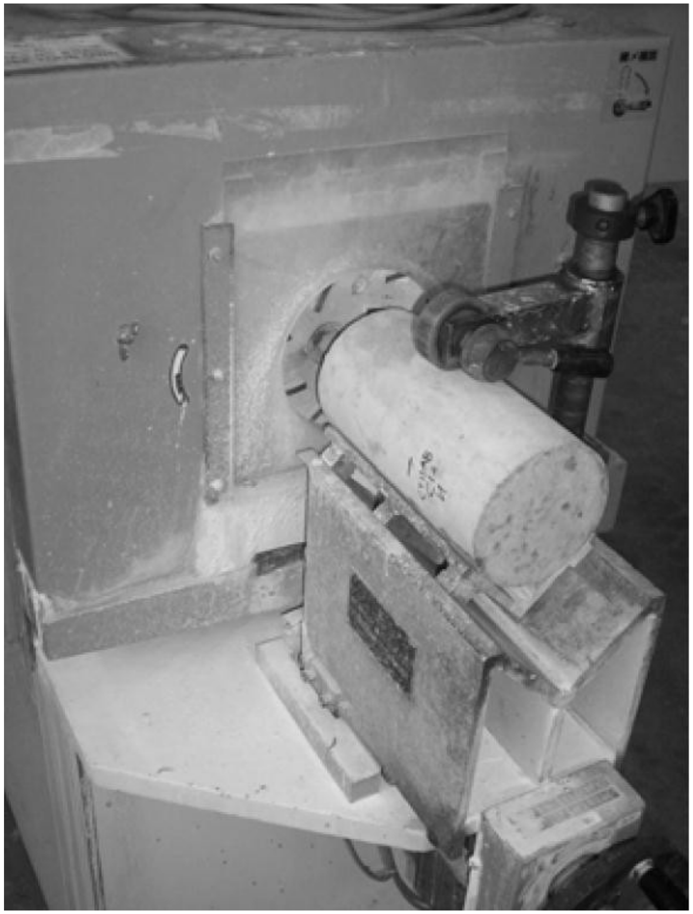
End Preparation

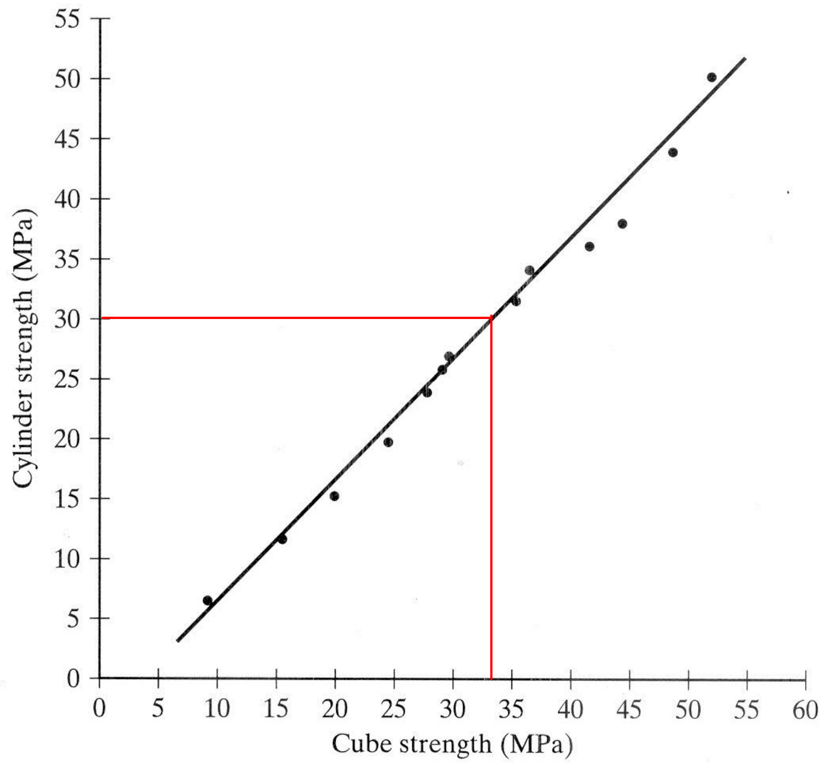
Capping



Grinding

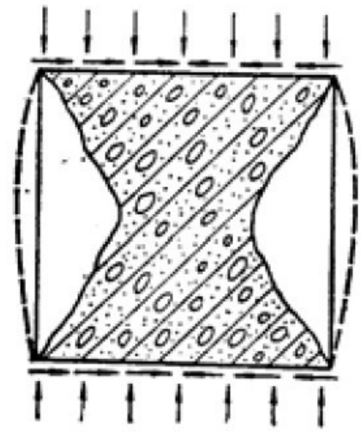
Cylinders vs. Cubes

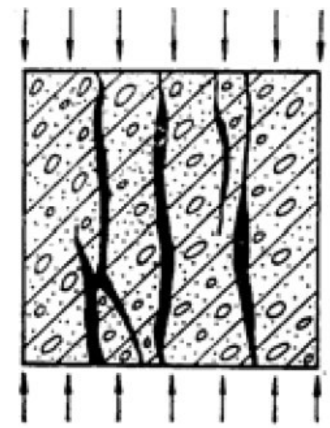
- Cubes do not need for end preparation
- Cube to cylinder strength ratio ranges from about 1.3 for low strength concrete to about 1.04 for higher strength concrete
  - Friction between the platens and the specimen ends creates much more confinement in cubes than in cylinders

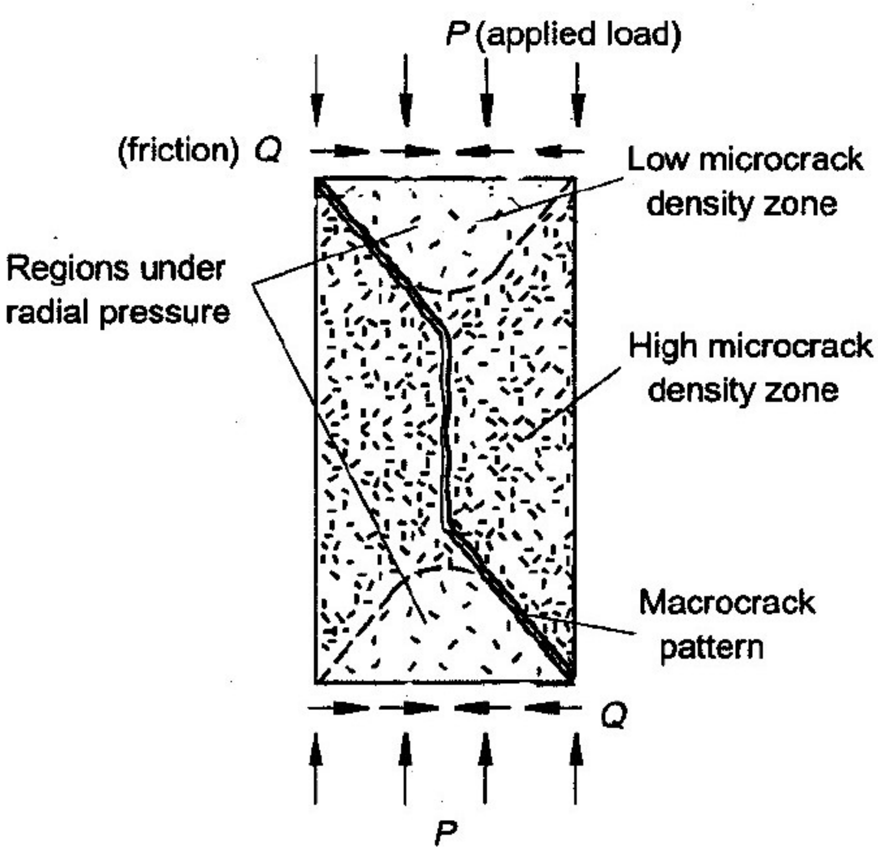


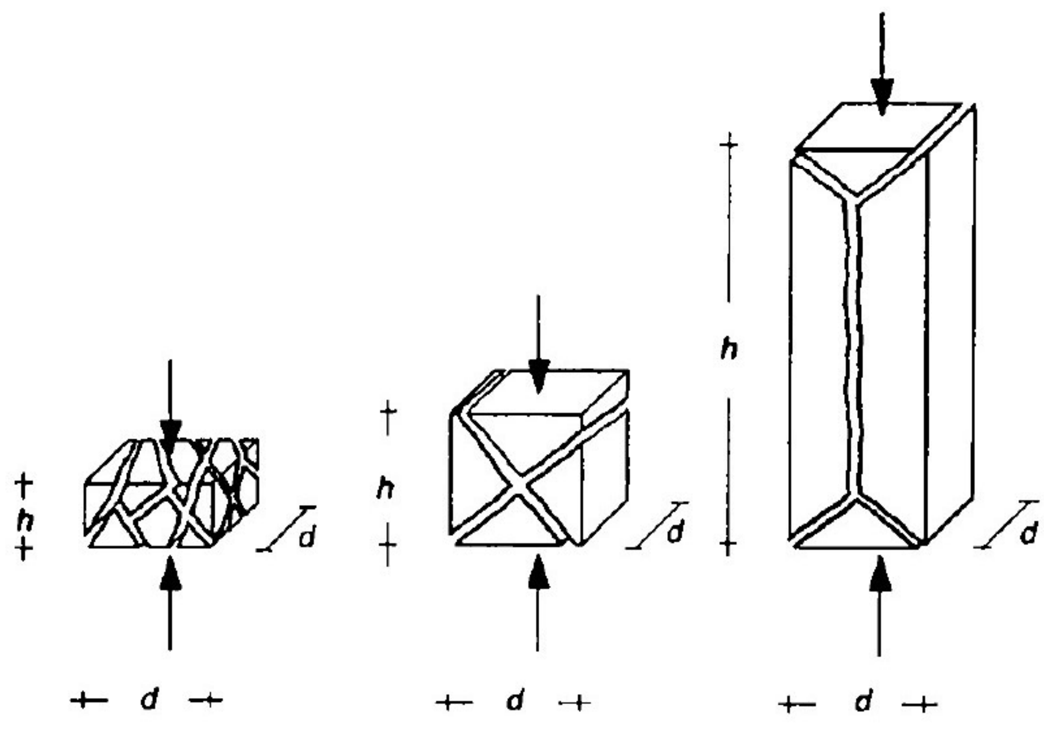
Failure Modes in Compression

- The confining pressure is greatest right at the specimen end and gradually dies out at a distance about $(\sqrt{3})\, d / 2 \approx 0.87\, d$
- For standard cylinder with $l/d = 2$, a small central portion of the cylinder is in true uniaxial compression



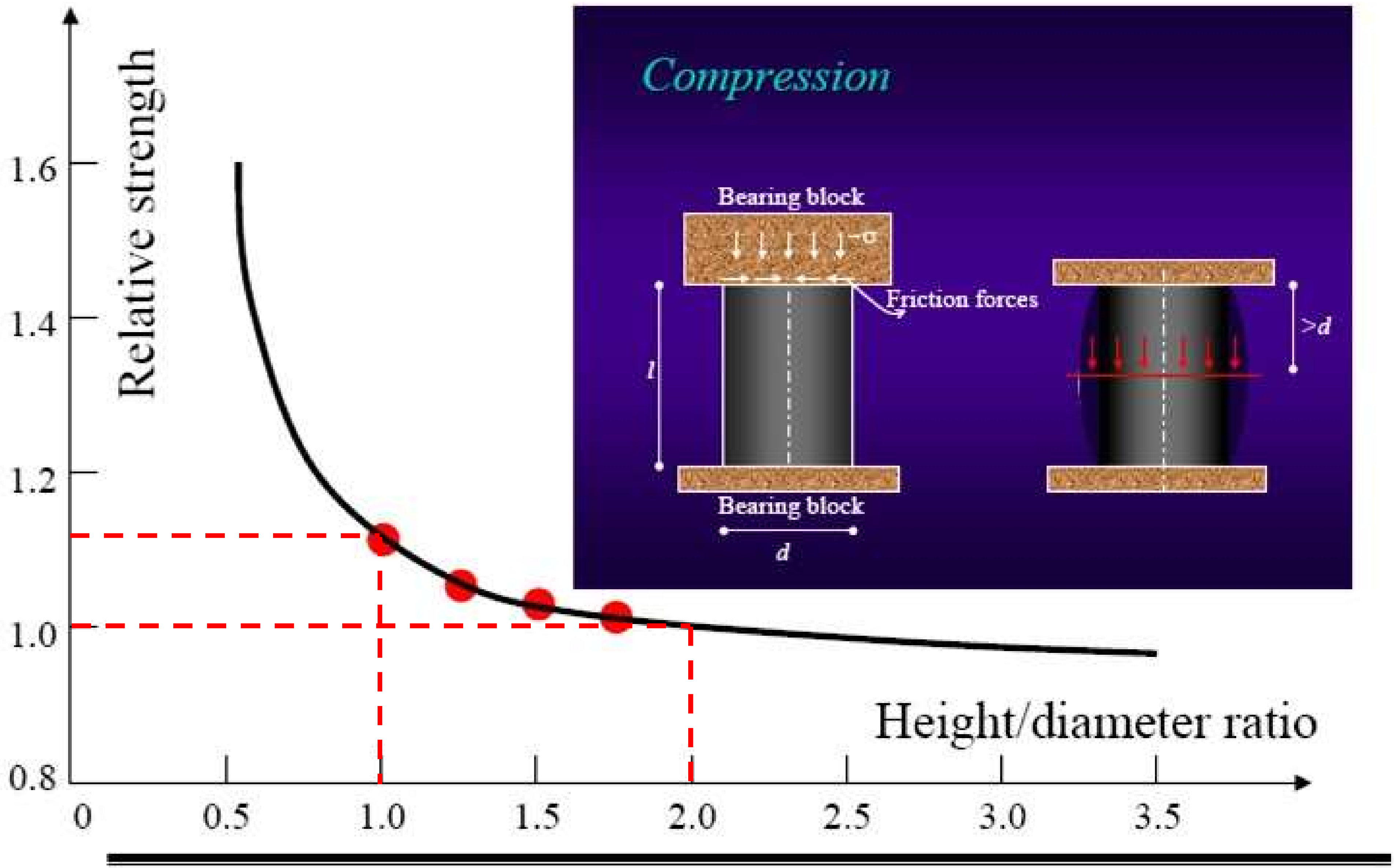






(a) (b) (c)

Effect of Height to Diameter Ratio on Cylinder Strength



Test on Tensile Strength

- Tensile strength is about 10% of compressive strength
- Direct tensile tests suffer from a number of difficulties related to
  - Holding the specimen properly in the testing machine without introducing stress concentration
  - The application of uniaxial tensile load which is free from eccentricity to the specimen
- Indirect methods

  Flexural tensile test

  Splitting tensile test

Flexural Tensile Test

- ❖ Standard test specimens
  - 150 x 150 x 750 mm
  - 100 x 100 x 500 mm
- ❖ Manner of loading
  - Third point loading
  - Central point loading
- ❖ Loading rate
  - 0.02 to 0.10 MPa/s

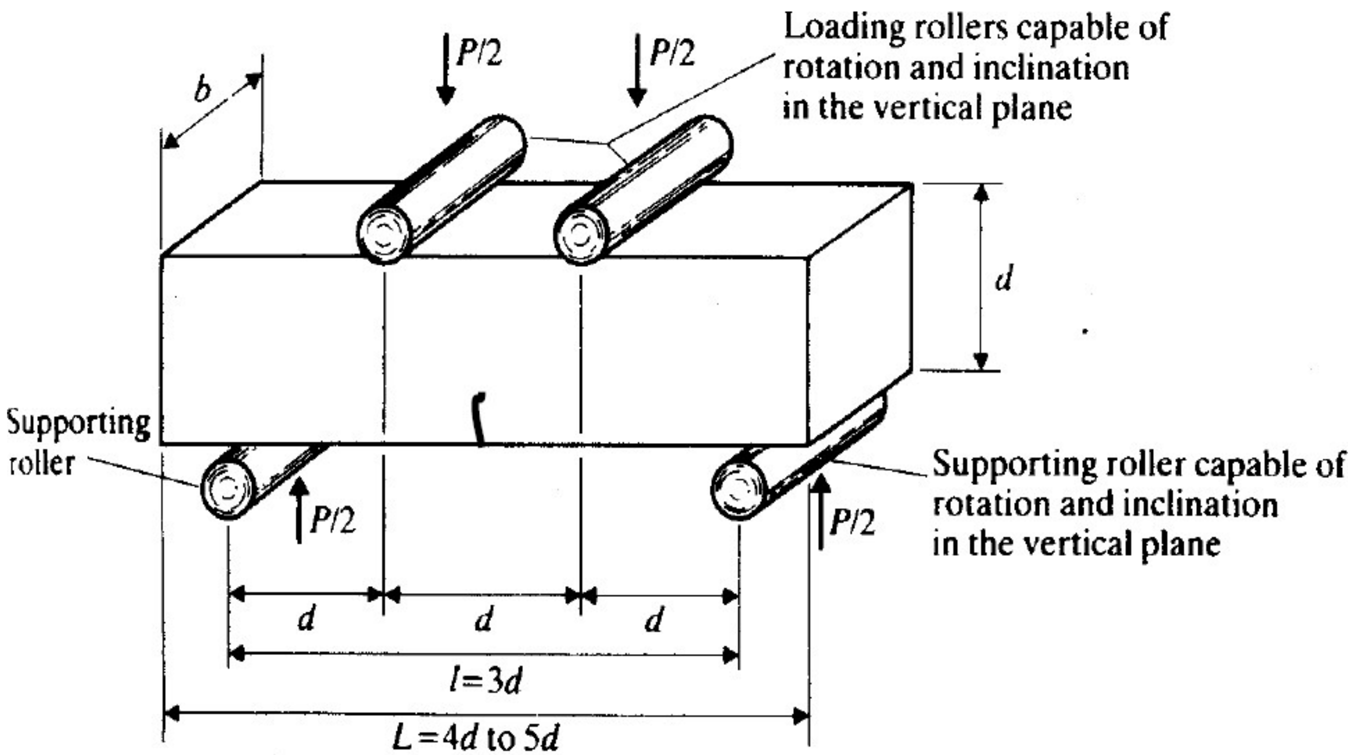
Flexural strength or modulus of rupture (MOR) is calculated

Third point flexure test

http://youtu.be/_U1ZIb4c5oA

Calculation of Flexural Strength

- $R = PL / bd^2$
  P: max. total load
  L: span length
  b: specimen width
  d: specimen depth



- Equation holds true only if the beam breaks between the two interior loading points

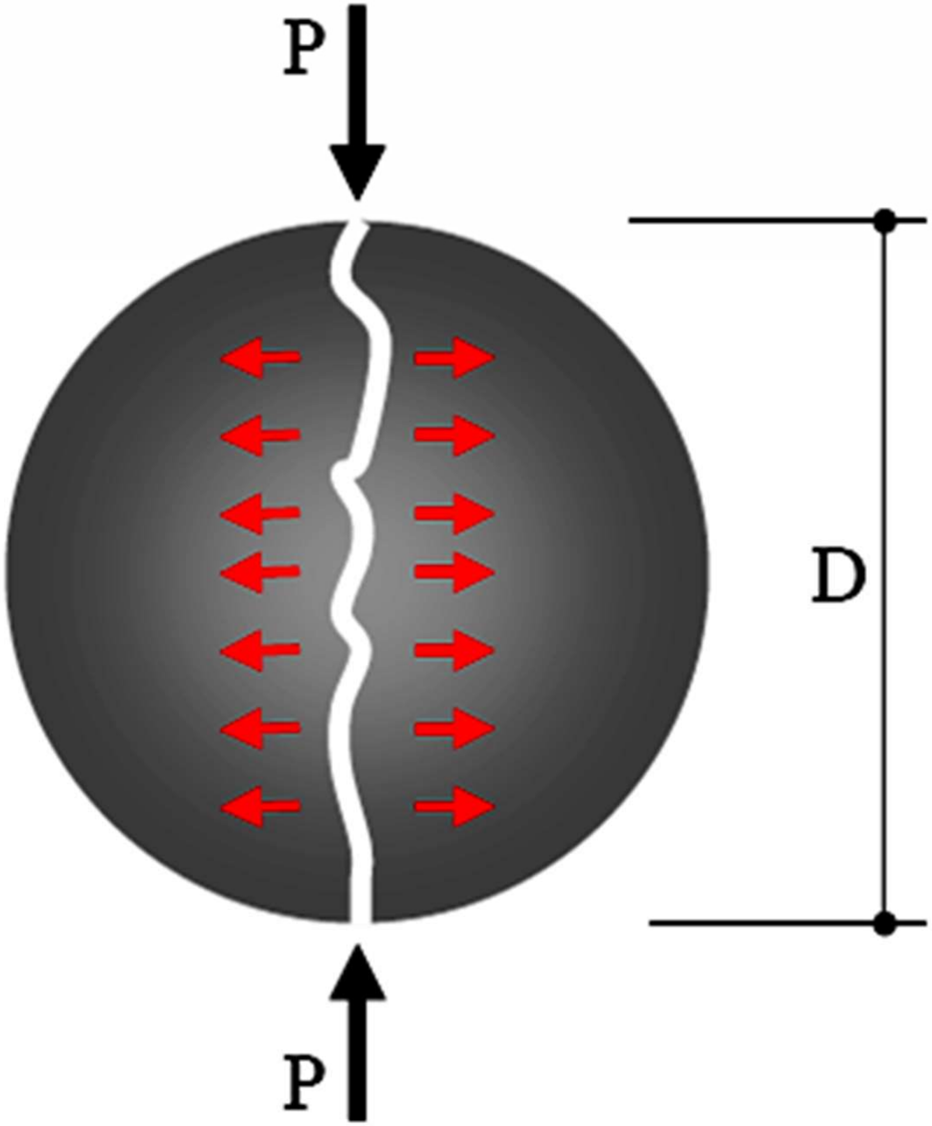
Splitting Tensile Test

- Standard test specimens
  - Cylinder
  - Cube
  - Prism
- Loading rate
  - 0.02 to 0.04 MPa/s

http://youtu.be/6lkZIrLp_mE

- Test is simple to perform and gives more uniform results than other tension tests
- Strength determined is closer to actual tensile strength of concrete
- Same molds can be used

Calculation of Splitting Tensile Strength



$$\sigma_{tensile} = \frac{2P}{\pi DL}$$

L = the length of the cylinder

Non-Destructive Tests (NDT)

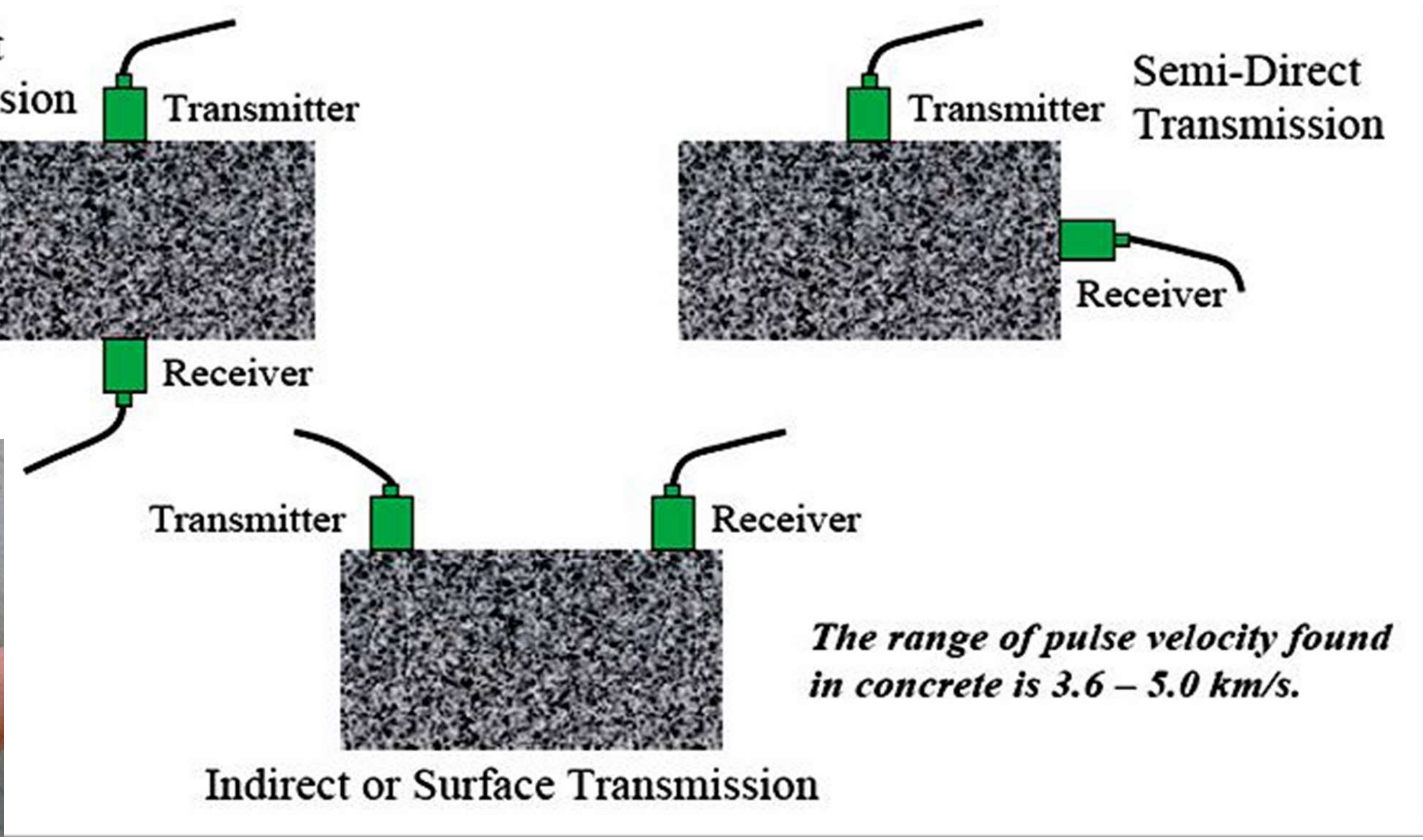
- **NDT** is a wide group of analysis techniques used in science and industry to evaluate the properties of a material, component or system without causing damage.
- Advantages
  - Speed
  - Cost
  - Lack of damage
  - Immediate availability of results
- Applications
  - New structures – QC
  - Existing structures – assessment of structural integrity or adequacy

Common NDT Techniques

- Ultrasonic pulse velocity test
- Rebound hammer/hardness test
- Windsor probe penetration resistance test
- Pull-out test

Ultrasonic Pulse Velocity Test

- Fundamental principle: Velocity of sound in solid material is a function of its elastic property

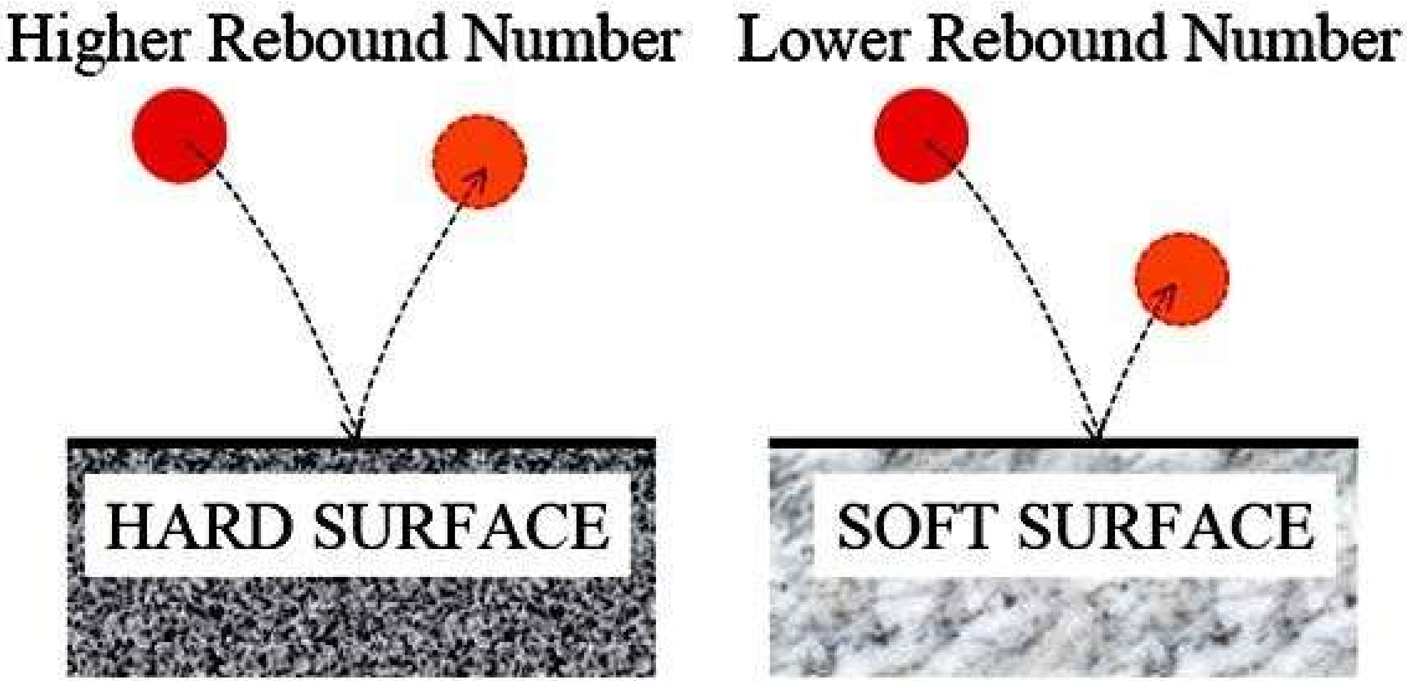


Ultrasonic Pulse Velocity Test

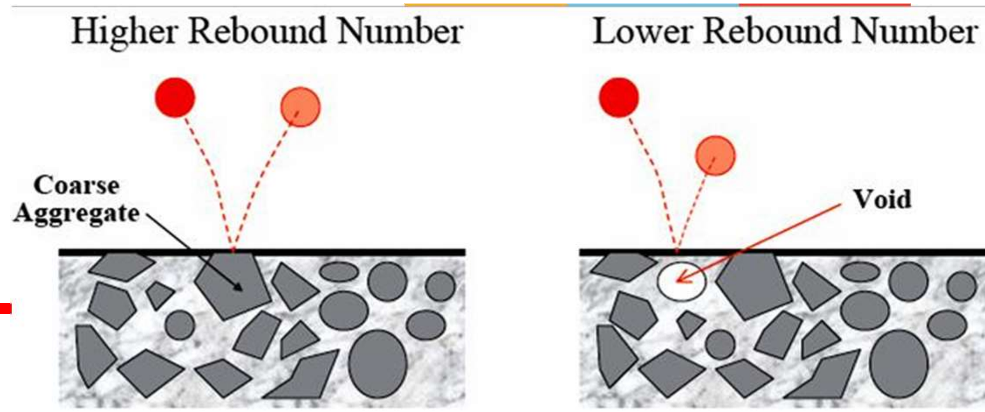
- No damage to concrete
- No simple correlation between compressive strength and pulse velocity, the correlation being affected by many factors
  - Aggregate type; Aggregate/cement ratio; Aggregate size and grading; Age of concrete; Curing conditions
- The ultrasonic pulse velocity test can give information about the interior of a concrete element and therefore useful to determine
  - Homogeneity of the concrete
  - Presence of voids, cracks or other imperfections.
  - Changes in a given concrete element (deterioration due to frost or fire)
  - Elastic modulus and dynamic Poisson's ratio of the concrete

Rebound Hammer/Hardness Test

- Fundamental principle: Rebound of elastic mass depends on surface hardness
- Surface hardness may be correlated to strength



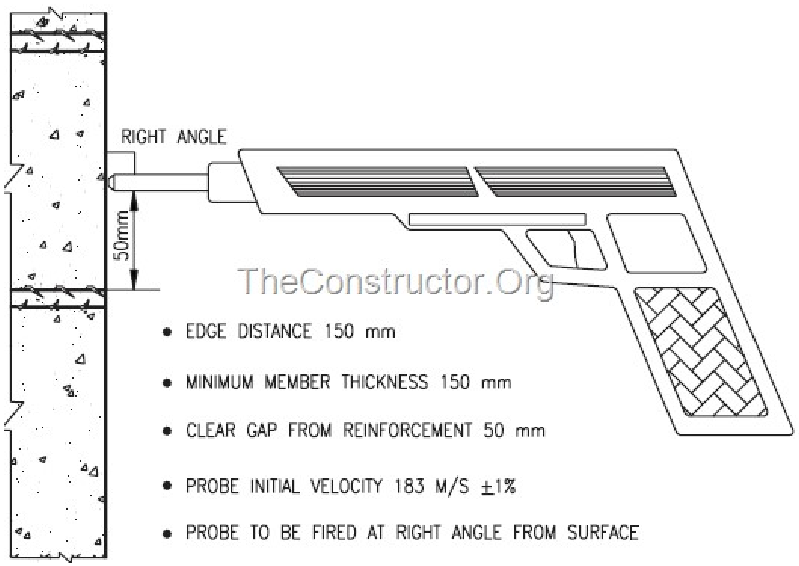
Rebound Hammer /Hardness Test

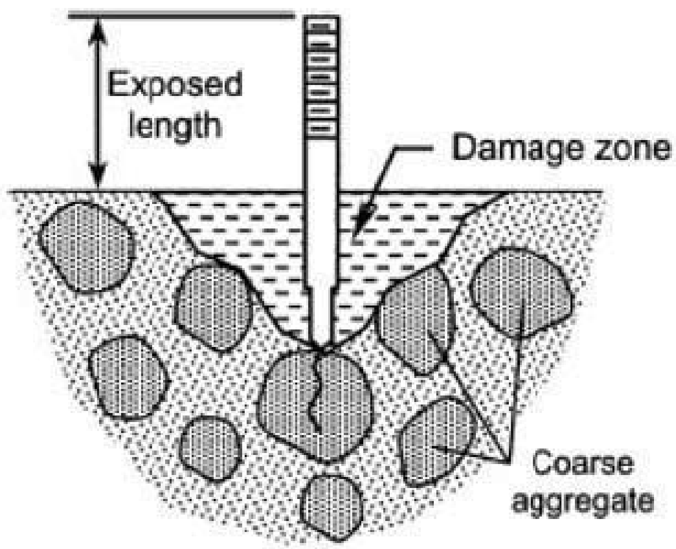


- No damage to concrete
- Measures properties of surface zone of concrete and therefore sensitive to local variations in concrete.
- Average of 25 readings from a 300 mm^2 area
- Correlation between hardness (rebound number) and strength of concrete is influenced by many factors
  - Surface finish; Carbonation; Moisture content of the concrete; Rigidity of member; Direction of impact
- The test method is useful for the following applications
  - Comparing a given concrete with a specified requirement
  - Checking the uniformity of concrete quality
  - Approximate estimation of strength
  - Abrasion resistance classification

Windsor Probe Penetration Resistance Test

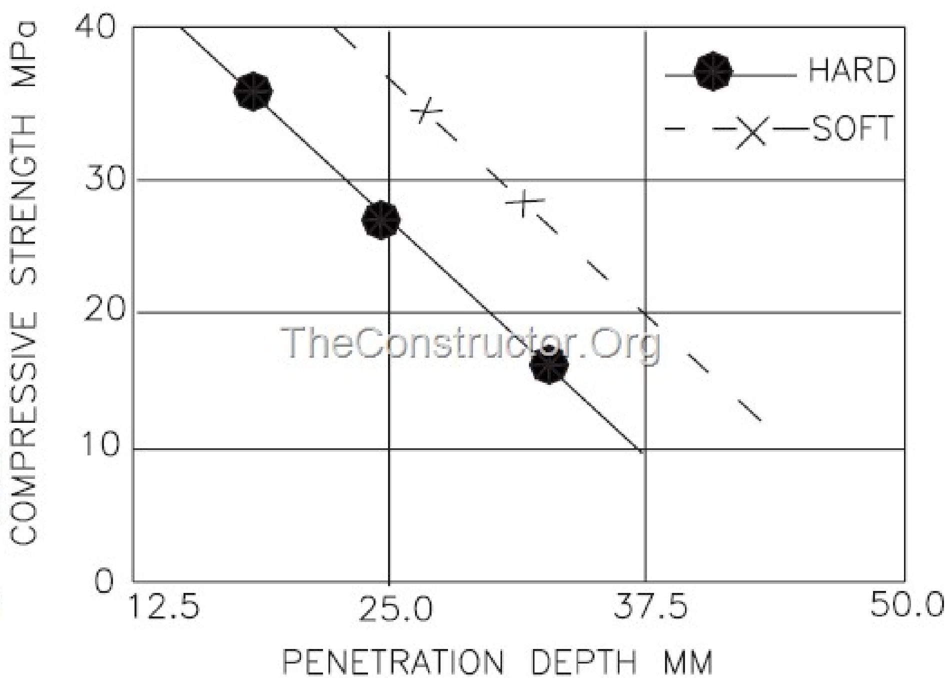
- Fundamental principle: Depth of penetration is inversely proportional to the compressive strength of concrete
- Minimal damage to concrete. Only creates small holes in the concrete surface.





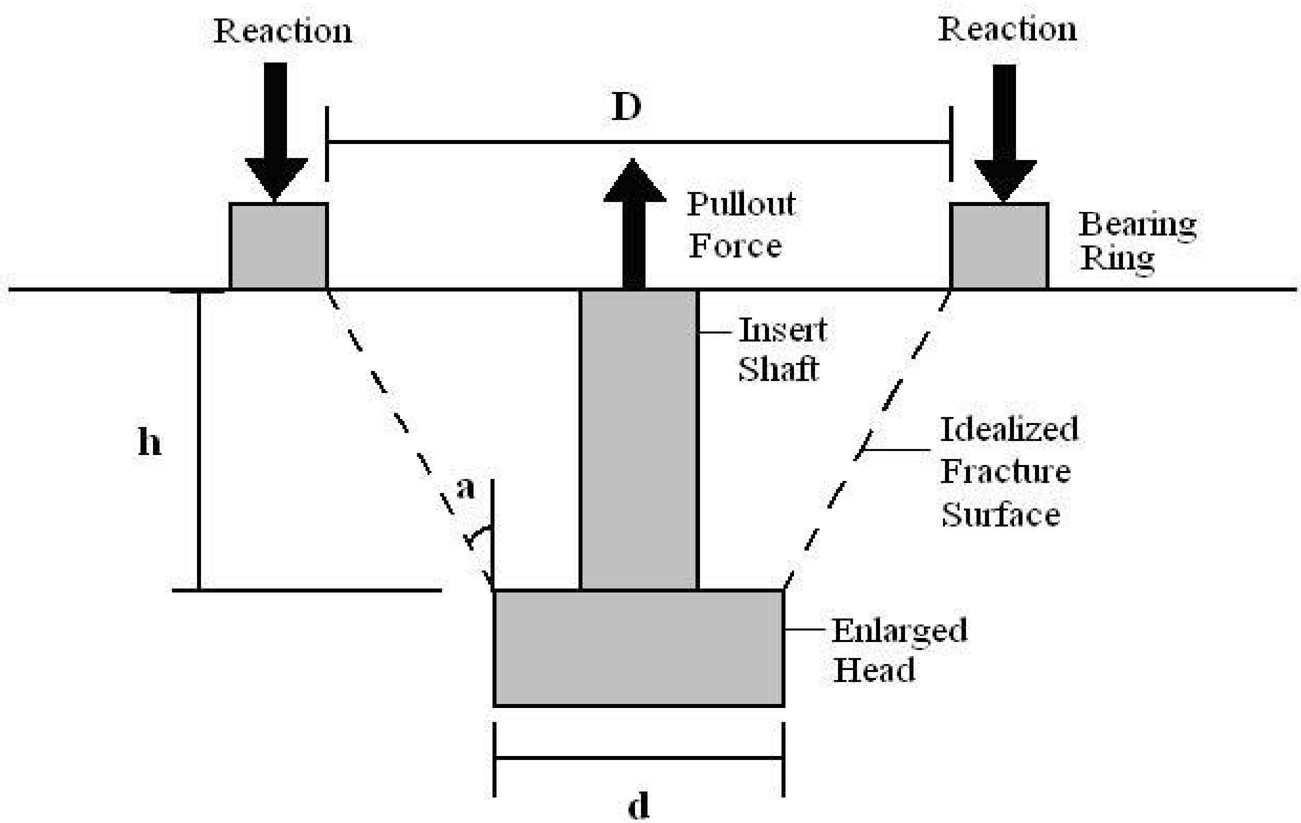
Windsor Probe Penetration Resistance Test

- Depth of penetration is sensitive to the hardness of aggregate
- Surface texture and carbonation have less effect than in the rebound hammer/hardness test since greater depth of concrete is tested
- Less no. of tests required than that of rebound hammer/ hardness test. Average of 3 tests in a triangular template.



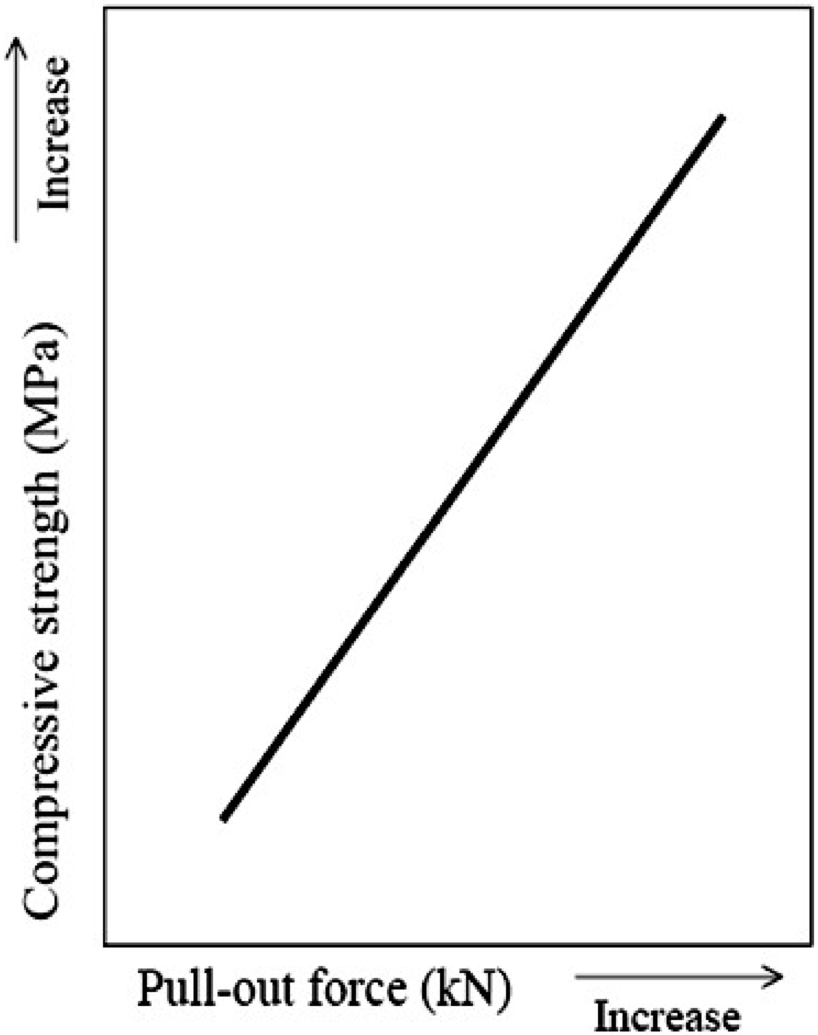
Pull-out Test

- Method: To measure the force required to pull out a previously cast-in steel rod with an embedded enlarged end
- Fundamental principle: Pull-out force is proportional to the compressive strength of concrete



Pull-out Test

- Pull-out force correlates well with the compressive strength of concrete for a wide range of curing conditions and ages.
- Concrete strength prediction is more precise than rebound hammer /hardness test and Windsor probe penetration test since a larger volume and a greater depth of concrete are tested.
- Greater damage to concrete. Repair of concrete is required.
- Need to plan beforehand



Core Tests

- Method: To conduct compression test on cores drilled from hardened concrete
- Enable the *visual inspection* of interior regions of the structural member along with the *estimation of actual concrete strength*
- Most precise prediction of concrete strength
- Greatest damage to concrete. Repair of concrete is necessary.

Comments on NDT

- NDTs do not measure concrete strength; rather, they provide an estimate of the concrete strength through correlation with some other property
- Fundamental shortcoming of all NDTs in which the property of concrete being measured is affected by various factors in a manner different from the influence of those factors on the strength of concrete

Patrick Cudahy Fire
Cudahy, Wisconsin, July 2009